5 **Estimation, confidence intervals and tests using a normal distribution**	5.1	Concepts of standard error, estimator, bias. Quality of estimators	The sample mean, $\bar{x}$, and the sample variance, $s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$, as unbiased estimates of the corresponding population parameters. Candidates will be expected to compare estimators and look for features such as unbiasedness and small variance.
	5.2	Concept of a confidence interval and its interpretation.	Link with hypothesis tests.
	5.3	Confidence limits for a Normal mean, with variance known.	Candidates will be expected to know how to apply the Normal distribution and use the standard error and obtain confidence intervals for the mean, rather than be concerned with any theoretical derivations.
	5.4	Hypothesis test for the difference between the means of two Normal distributions with variances known.	Use of $\dfrac{(\bar{X}-\bar{Y})-(\mu_x-\mu_y)}{\sqrt{\dfrac{\sigma_x^2}{n_x}+\dfrac{\sigma_y^2}{n_y}}} \sim \mathrm{N}(0,\,1)$

Topics	What students need to learn:		
	Content		Guidance
5 **Estimation, confidence intervals and tests using a normal distribution** *continued*	5.5	Use of large sample results to extend to the case in which the population variances are unknown.	Use of Central Limit Theorem and use of $\dfrac{(\overline{X}-\overline{Y})-(\mu_x-\mu_y)}{\sqrt{\dfrac{S_x^2}{n_x}+\dfrac{S_y^2}{n_y}}} \approx\sim \mathrm{N}(0,1)$